Relative price dispersion and inflation regimes in South Africa

Keagile Lesame[*] Xolani Sibande[†]

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[*]Corresponding author. National Treasury, South Africa. Email: Keagile.Lesame@treasury.gov.za
[†]Economic Research Department, South African Reserve Bank. Email: xolani.sibande@resbank.co.za.

1 Introduction

Inflation is known to have adverse welfare effects through its distortion of relative prices which causes the misallocation of resources, production inefficiencies leading to lower real economic activity. This is well supported by both theoretical and empirical studies. This paper contributes to the empirical literature, by examining the relationship between inflation and relative price variability (RPD) in South Africa given the little evidence available and to better understand how this relationship has evolved and the implications for monetary policy.

The South African context can provide further insights into the relationship between inflation and RPD given the shifts in the inflation targets from 3-6 per cent in 2000 to 4.5 per cent since 2017 as structural changes in inflation can affect the link between inflation and relative price variability (Caglayan and Filiztekin, 2003) and that the functional form of relationship is U-shaped and dynamic (Fielding and Mizen, 2000; Choi, 2010). The latter motivates the estimation of the inflation rate that minimizes relative price distortions since if the relationship was linear then the optimal inflation rate would be zero. Given the inflation target reform towards a 3% in South Africa, this study contributes to the policy discussion on the appropriate inflation target and understating how dynamic relations between inflation and relative price variability has evolved.

The theoretical literature generally distinguishes between menu-cost and signal extraction models in explaining the positive relation between trend inflation and RPD. In these models' firms follow a one-sided dynamic pricing rule when setting prices and face fixed non-zero costs of adjusting nominal prices (e.g. printing new menus) when inflation increases as the latter reduces the optimal real price of goods and services. Due to differences in administrative costs, these nominal price adjustment costs vary across firms moreover these adjustments occur at different times which implies staggard price setting behaviour which leads to relative price variability. Menu cost models predict a that higher expected inflation causes an increase in RPD (citations xxx) while Lucas-type signal extraction models (Lucas, 1973; Hercowitz, 1981; Cukierman, 1983) argue unexpected inflation uncertainty causes higher RPD because informational frictions make it difficult to distinguish between idiosyncratic relative shocks or an aggregate (inflationary) shock. Menu cost and signal-extraction models assume a positive and linear relationship making the optimal inflation rate zero, however, recently monetary search models (Head and Kumar, 2005) show that this relationship is nonlinear making the optimal inflation rate somewhere between zero and positive inflation rate.

2 Citations

Adam et al. (2024) Baglan et al. (2016) Ball and Romer (1993) Benabou and Gertner (1993) Caglayan and Filiztekin (2003) Caglayan et al. (2008) Grier and Perry (1996) Lach and Tsiddon (1992) Nakamura et al. (2018) Ndou and Redford (2014) Ndou and Gumata (2017) Parks (1978) Parsley (1996) Rather et al. (2018)

3 Understanding relative price dispersion through Lucas's aggregate supply curve

$$y_t(z) = y_{n,t} + y_{ct}(z),$$

where $y_t(z)$ is the aggregate supply at time t in market z, $y_{n,t}$ is the natural or secular level of supply at time t, and y_{ct} is the cyclical component of supply at time t.

$$y_{n,t} = \alpha + \beta t,$$

where α is a constant and β captures the trend growth rate of the natural level of supply.

$$y_{ct}(z) = \gamma[P_t(z) - E(P_t|I_t(z))] + \lambda y_{c,t-1}(z),$$

where γ captures the sensitivity of supply to unexpected price changes, $P_t(z)$ is the actual price level in market z at time t, $E(P_t|I_t(z))$ is the expected price level based on information available at time t in market z, and λ captures the persistence of the cyclical component and $|\lambda| < 1$. $E(P_t|I_t(z))$ is current expected price level conditioned on information that is available at time t, at market z.

$I_t(z)$

$$P_t \sim N(\overline{P_t}, \sigma^2),$$

$$z \overset{\text{iid}}{\sim} N(0, \tau^2)$$

$$P_t(z) = P_t + z$$

$$E(P_t | I_t(z)) = E(P_t | P_t(z), \overline{P_t}) = (1-\theta)P_t(z) + \theta\overline{Pt}$$

$$\theta = \frac{\tau^2}{(\sigma^2 + \tau^2)}$$

$$y_t(z) = y_{nt} + \theta\gamma[P_t(z) - \overline{P_t}] + \lambda y_{c,t-1}$$

$$y_t = y_{nt} + \theta\gamma(P_t - \overline{P_t}) + \lambda[y_{t-1} - y_{n,t-1}]$$

$\theta \uparrow$ in τ^2 and $\downarrow$ in σ^2. γ

4 Empirical literaturer review

5 Data and methodology

Our analysis uses monthly inflation data on 45 goods categories from January 2009 to NoV 2025 obtained from Statistics South Africa (Stats SA).[1] . Figure A1 below shows the long term inflation rate in South Africa, and Figure A2 in the appendix shows the various price categories used in the analysis. Table A1 provides the descriptive statistics of the inflation rates for the various price categories used in the analysis.

[1]The data is available at: http://www.statssa.gov.za/ The data is seasonally adjusted.

The relative price dispersion (RPD) is calculated in line with Lach and Tsiddon (1992) as the weighted standard deviation of individual price inflation rates from the aggregate inflation rate. The RPD is essentially the variability of relative prices of a given product. We first calculate the monthly inflation rate as follows:

$$\pi_{i,t} = ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right) * 100, \tag{1}$$

where $\pi_{i,t}$ is the inflation rate of good i at time t, $P_{i,t}$ is the price of good i at time t, and $P_{i,t-1}$ is the price of good i at time $t-1$. Then, the RPD is calculated as:

$$RPD_t = \sqrt{\sum_{i=1}^{n} \omega_i (\pi_{i,t} - \pi_t)^2}, \tag{2}$$

where ω_j is the expenditure share of the j^{th} good, π_t is the headline or aggregate inflation rate at time t, and $\sum_{i=1}^{n} \omega_i = 1$. The expenditure weights (ω_i) used are derived from the Consumer Price Index (CPI) basket provided by Stats SA. Figure 1 shows the RPD calculated using the above formula. We use the 2025 CPI Stats SA weights. The weights are shown in Figure A3 in the appendix.

To examine the relationship between inflation and RPD, we estimate the following econometric model:

$$RPD_t = \beta_0 + \beta_1 |\pi_t| + \sum_{h=1}^{3} \omega_h RPD_{t-h} + \mu_t, \tag{3}$$

where RPD_t is the relative price variance at time t, $|\pi_t|$ is the absolute value of aggregate inflation rate at time t, RPD_{t-h} are lagged values of RPD to account for its dynamic nature, and μ_t is the error term. The coefficient β_1 captures the effect of inflation on RPD. We include three lags of RPD based on Akaike Information Criterion (AIC) (and other tests) and to capture potential persistence in RPD. For robustness, we repeat this estimation using core inflation (excluding food and non-alcoholic beverages, and fuel and electricity) instead of headline inflation to account for the potential volatility in these components.

The literature on RPD emphasises that the relationship between inflation and RPD maybe time-varying due to structural changes in the economy, monetary policy regimes, and other factors. To capture potential time variation in the relationship, we also estimate a rolling regression model with a 24-month and 60-month windows as follows:

$$RPD_t = \alpha_t + \beta_{1,t}|\pi_t| + \sum_{h=1}^{3} \omega_{h,t} RPD_{t-h} + \mu_t, \tag{4}$$

where the coefficients α_t, $\beta_{1,t}$, and $\omega_{h,t}$ vary over time. This approach allows us to observe the stability of the coefficients over time, which is an indicator of non-linearity in the relationship between RPD and inflation.

Next, we utilise a semi-parametric or partially linear approach to estimate the optimal level of inflation given the RPD. This approach allows us to model the relationship between inflation and RPD without imposing a strict functional form. Following the literature, we first specify a quadratic (U-shaped) relationship between inflation and RPD as follows:

$$RPD_t = \alpha + \pi_t + \beta_2 \pi_t^2 + \sum_{h=1}^{3} \phi_h RPD_{t-h} + \epsilon_t, \tag{5}$$

where β_2 captures the curvature of the relationship between inflation and RPD. The optimal inflation rate that minimizes RPD can be derived by taking the first derivative of the equation with respect to π_t and setting it to zero, yielding:

$$\pi_t^* = -\frac{1}{2\beta_2}. \tag{6}$$

To estimate this optiomal inflation we implement a semi parametric approach as follows:

$$RPD_t = f(\pi_t) + \sum_{h=1}^{3} \phi_h RPD_{t-h} + \epsilon_t, \tag{7}$$

where $f(\pi_t)$ is a non-linear smooth differentiable function of inflation and the RPD. Therefore, the optimal inflation rate can be obtained by finding the minimum of the estimated $f(\pi_t)$ function. $f(\pi_t)$ is estimated in two stages. First we estimate λ_h as follows:

$$RPD_t = \sum_{h=1}^{3} \lambda_h \overline{RPD}_{t-h} + \eta_t, \tag{8}$$

where $\overline{RPD}_{t-h}$ are the residuals from a non-parametric regression of each lag of RPD_t on π_t. In the second stage, we estimate $f(\pi_t)$ non-parametrically as follows:

$$\hat{\eta}_t = f(\pi_t) + v_t, \tag{9}$$

where $\hat{\eta}_t$ are the residuals from Equation 8, and $f(\pi_t)$ is estimated using a kernel regression method. The bandwidth (h) for the kernel estimator is selected using cross-validation to minimize the mean squared error. As a treatment for extreme values of inflation, we further implement the unbounded Gaussian kernel and the outlier-robust Epanechnikov kernel. In essence, the semi-parametric approach captures the sensitivity of RPD to marginal increases in inflation. In that case, the optimal inflation rate can be obtained by finding the point where the first derivative ($f'(\pi_t)$) is equal to zero. Therefore, if $f'(\pi_t) > 0$ ($f'(\pi_t) < 0$), then an increase in inflation (decrease) leads to an increase (decrease) in the RPD. For robustness we estimate the $g(UIN_t)$ with multiple bandwith (h).

To further disentangle the effects of RPD on inflation, we decompose headline inflation into its expected and unexpected components. As discussed in the literature section, for example, theory suggests that inflation uncertainty has a more pronounced effect on RPD due to information frictions faced by firms when adjusting prices (Lucas, 1973; Hercowitz, 1981; Cukierman, 1983).

We therefore estimate the following model:

$$RPD_t = \alpha + \beta_0 EIN_t + \beta_1 UIN_t + g(UIN_t) + \sum_{k=1}^{3} \lambda_k RPD_{t-k} + \epsilon_t, \tag{10}$$

where EIN_t is the expected inflation rate, UIN_t is the unexpected inflation rate, and $g(UIN_t)$ is a non-linear function of unexpected inflation (UIN_t). We obtain the expected using **a simple moving average of headline inflation, and unexpected inflation as the standard deviation of inflation**. The estimation of $g(UIN_t)$ follows the same two-stage semi-parametric approach outlined in Equation 8 and Equation 9. The number of lags is the same as in Equation 3.

Lastly, and some-what novel to the RPD literature, we employ a quantile on quantile regression approach (Koenker, 2017). Quantile on quantile regressions are robust to outliers and allows us to estimate the regime aspects of the relationship between RPD and inflation. For brevity, we explain the quantile on quantile estimation procedure in the Appendix.

6 Results

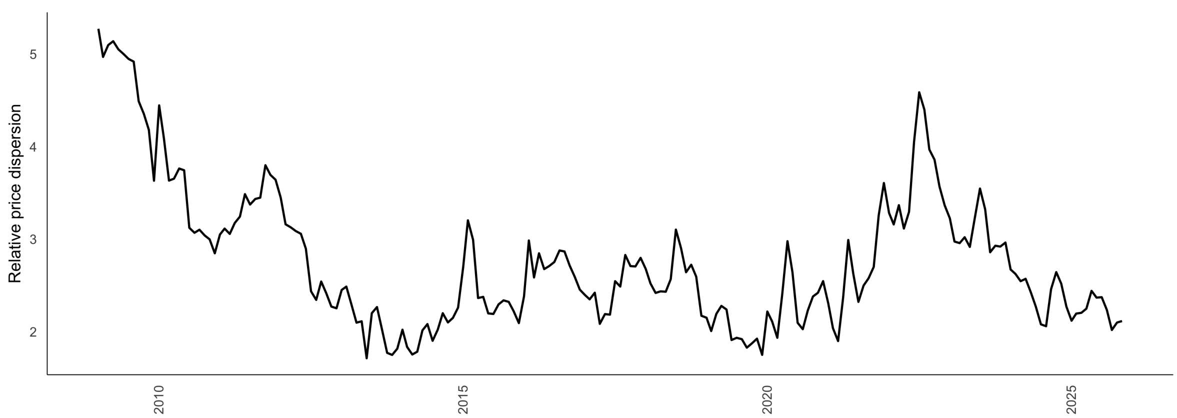

Figure 1: RPD over-time

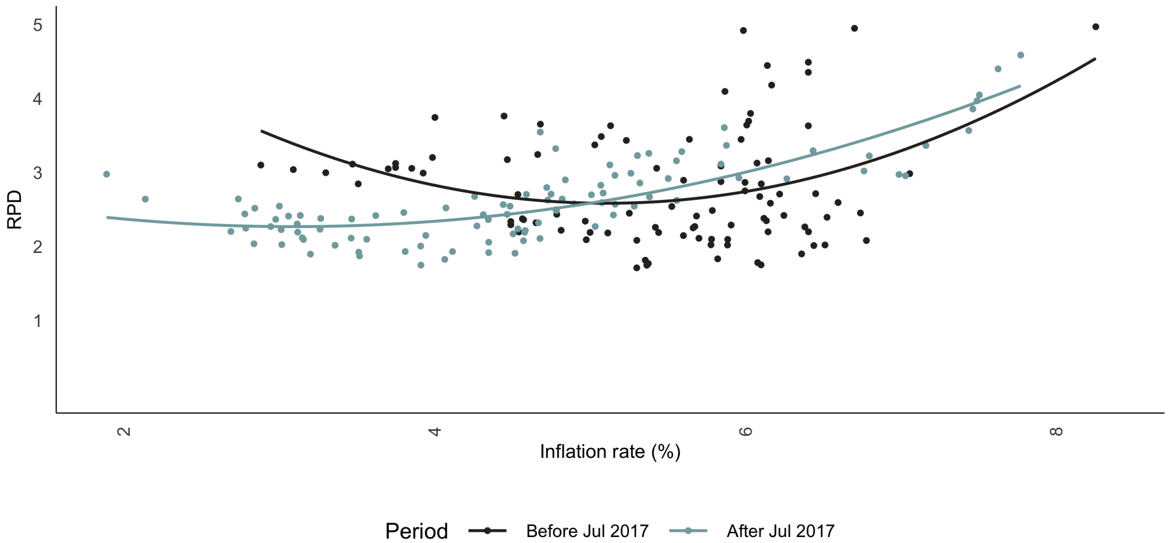

Figure 2: RPD vs headline Inflation

Table 1: Baseline regressions

	Full Sample	Pre 2017	Post 2017	Full Sample: Squared Inflation	Pre 2017: Squared Inflation	Post 2017: Squared Inflation
π_t	0.01*	0.009	0.032***	-0.046	-0.104	-0.033
π_t^2				0.006*	0.011*	0.007*
RPD_{t-3}	0.244***	0.173*	0.248**	0.234***	0.154	0.209**
RPD_{t-2}	-0.382***	-0.14	-0.645***	-0.363***	-0.111	-0.609***
RPD_{t-1}	1.039***	0.896***	1.116***	1.017***	0.864***	1.063***
D_{2022m8}	0.026		0.024	-0.001		-0.012
D_{2020m2}	-0.116		-0.213**	-0.106		-0.205**
D_{2017m7}	0.156*			0.161*		
D_{2014m7}	-0.098	-0.087		-0.105	-0.096	
$D_{2011m12}$	0.042	0.027		0.047	0.037	
C	0.044	0.012	0.117***	0.184**	0.326*	0.304**

Note: *** = p value < 0.01, ** = 0.01 < p value < 0.05, * = 0.05 < p value < 0.1.

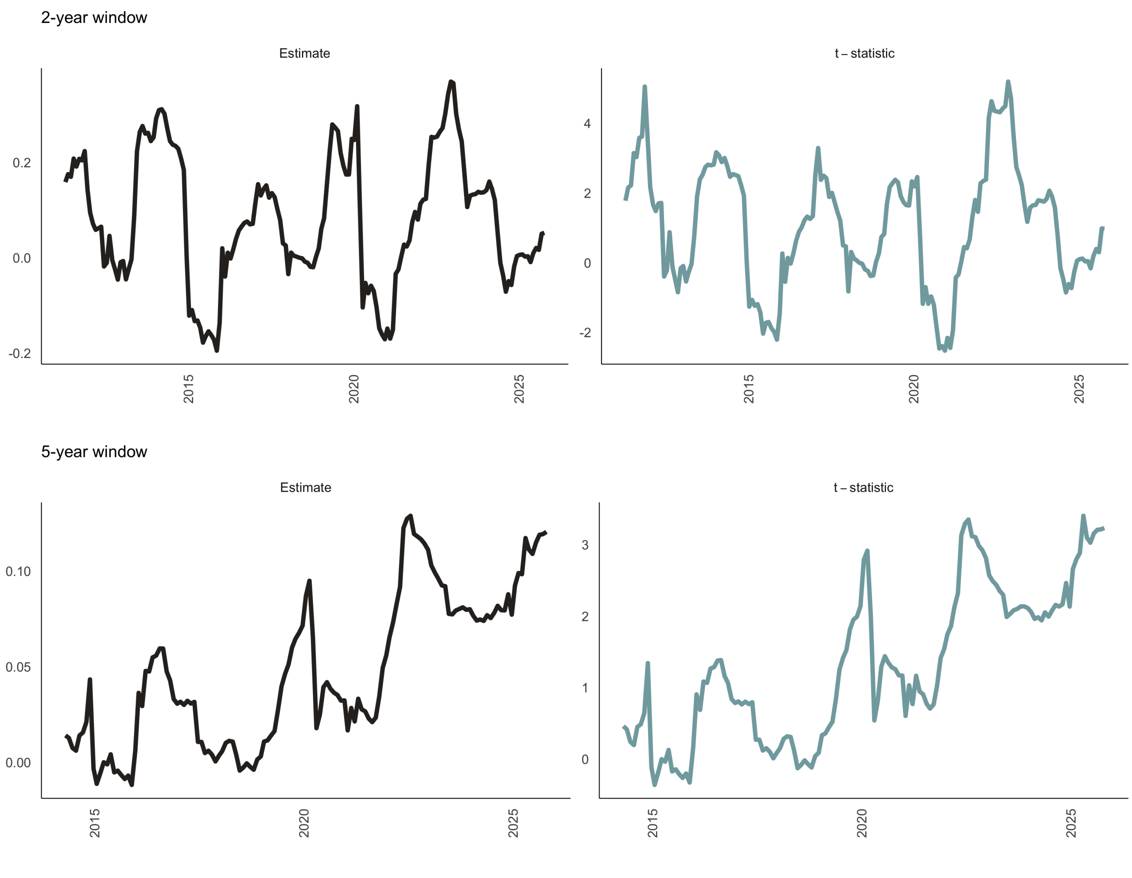

Figure 3: Rolling regressions. Note: t-statistics of greater (or less) than 1.96 (or -1.96) is sigficant at a 5% level of significance.

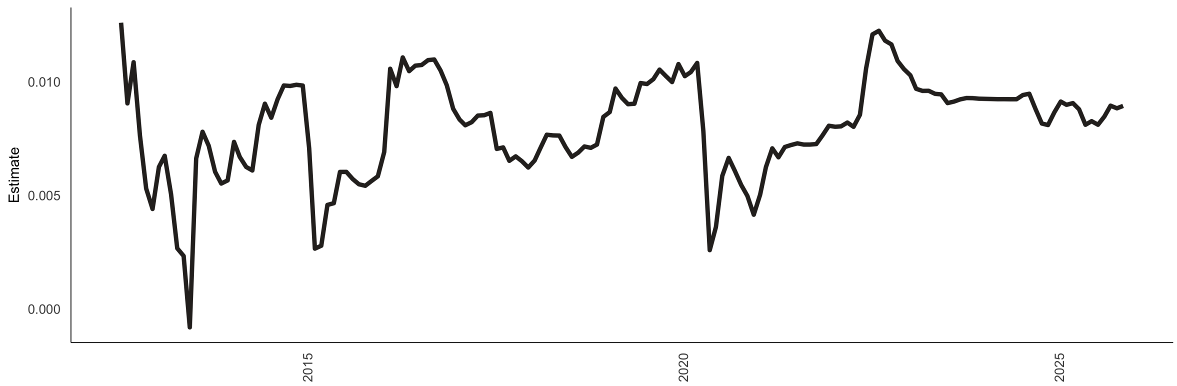

Figure 4: Recursive regressions. Note: Recursive regression estimates of the coefficient on absolute headline inflation with a fixed start date of 2012-07.

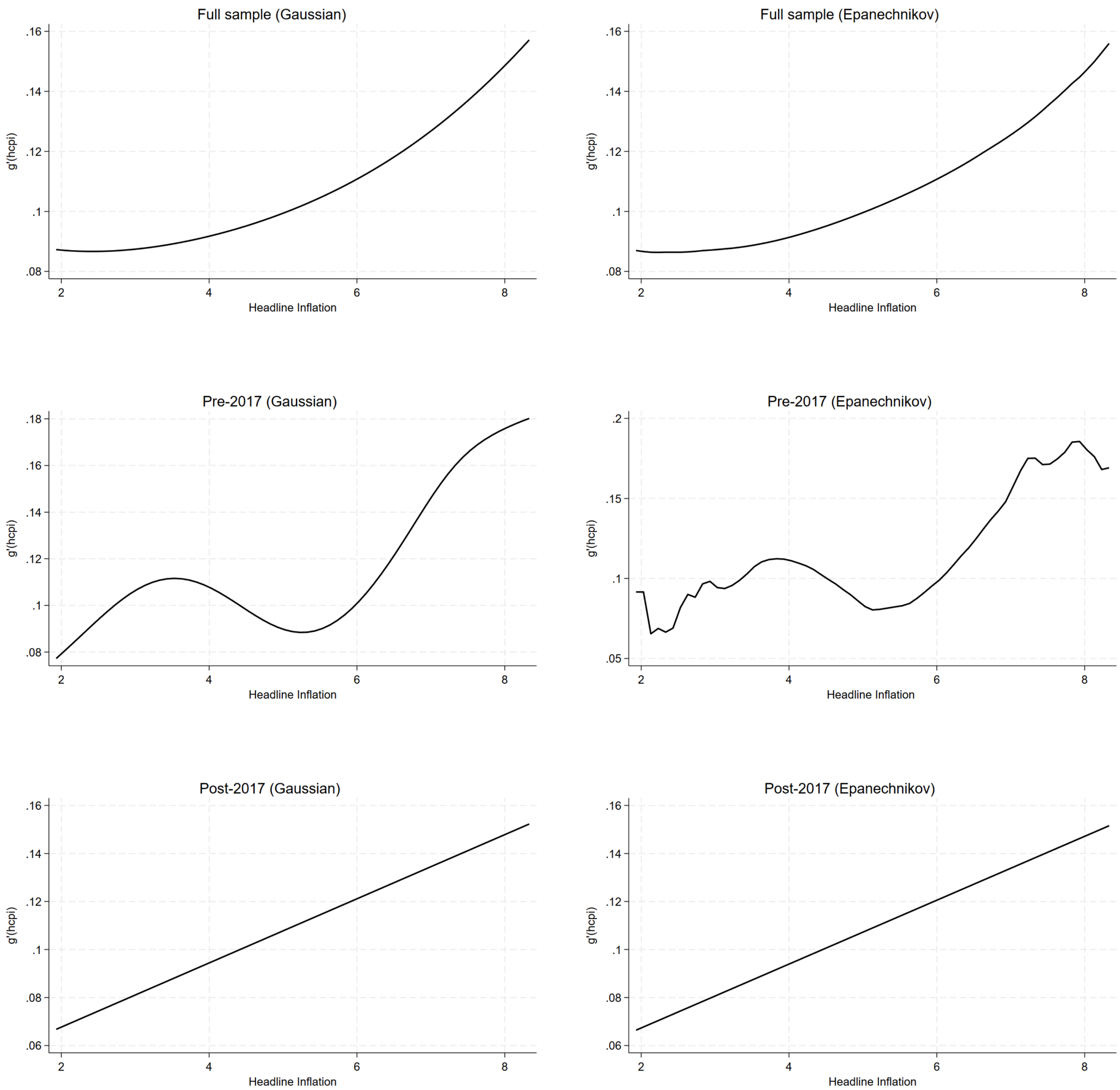

Figure 5: Semi-parametric regressions

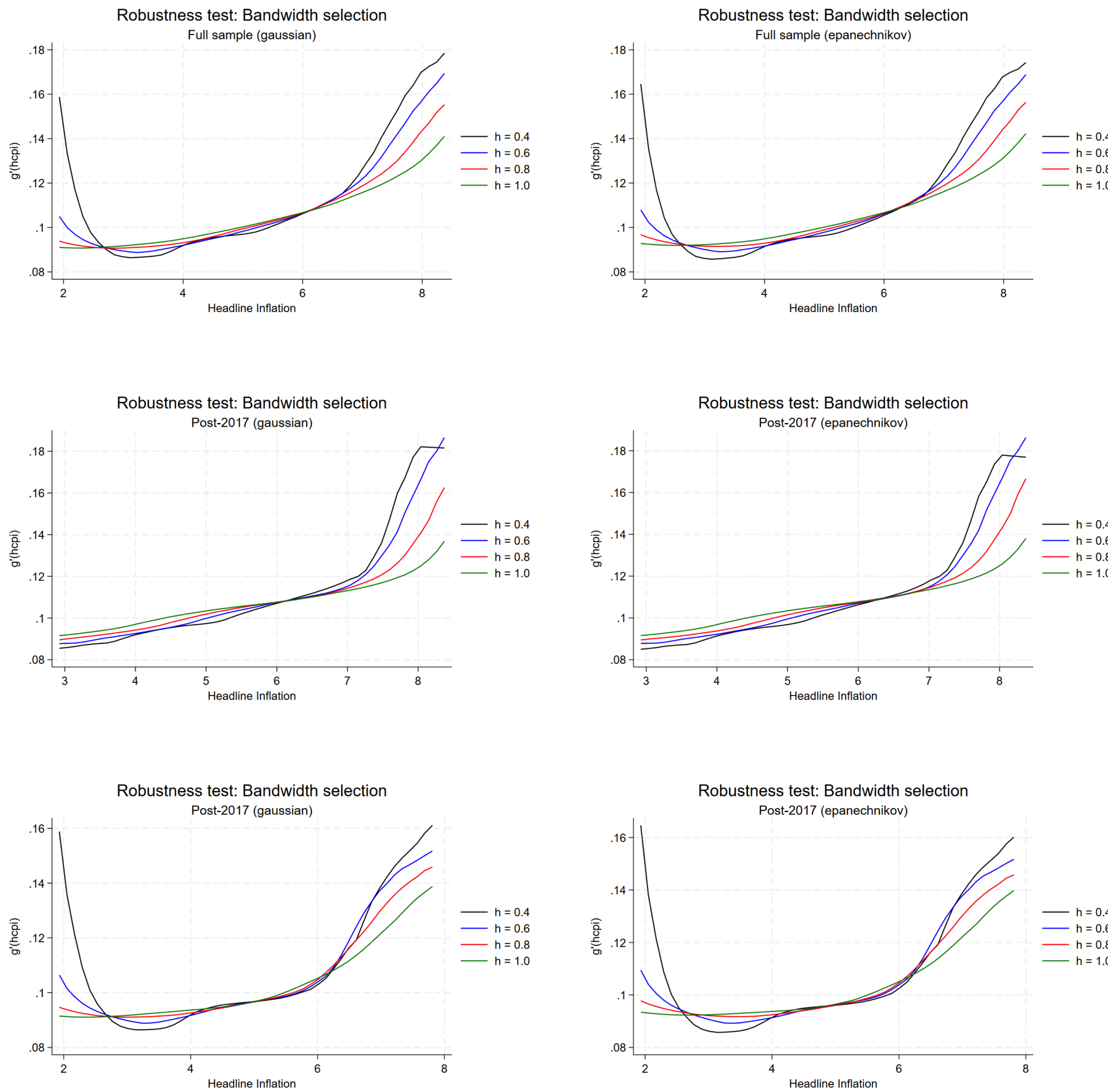

Figure 6: Semi-parametric regressions with alternative bandwidth selection. Note: hcpi is the headline inflation rate.

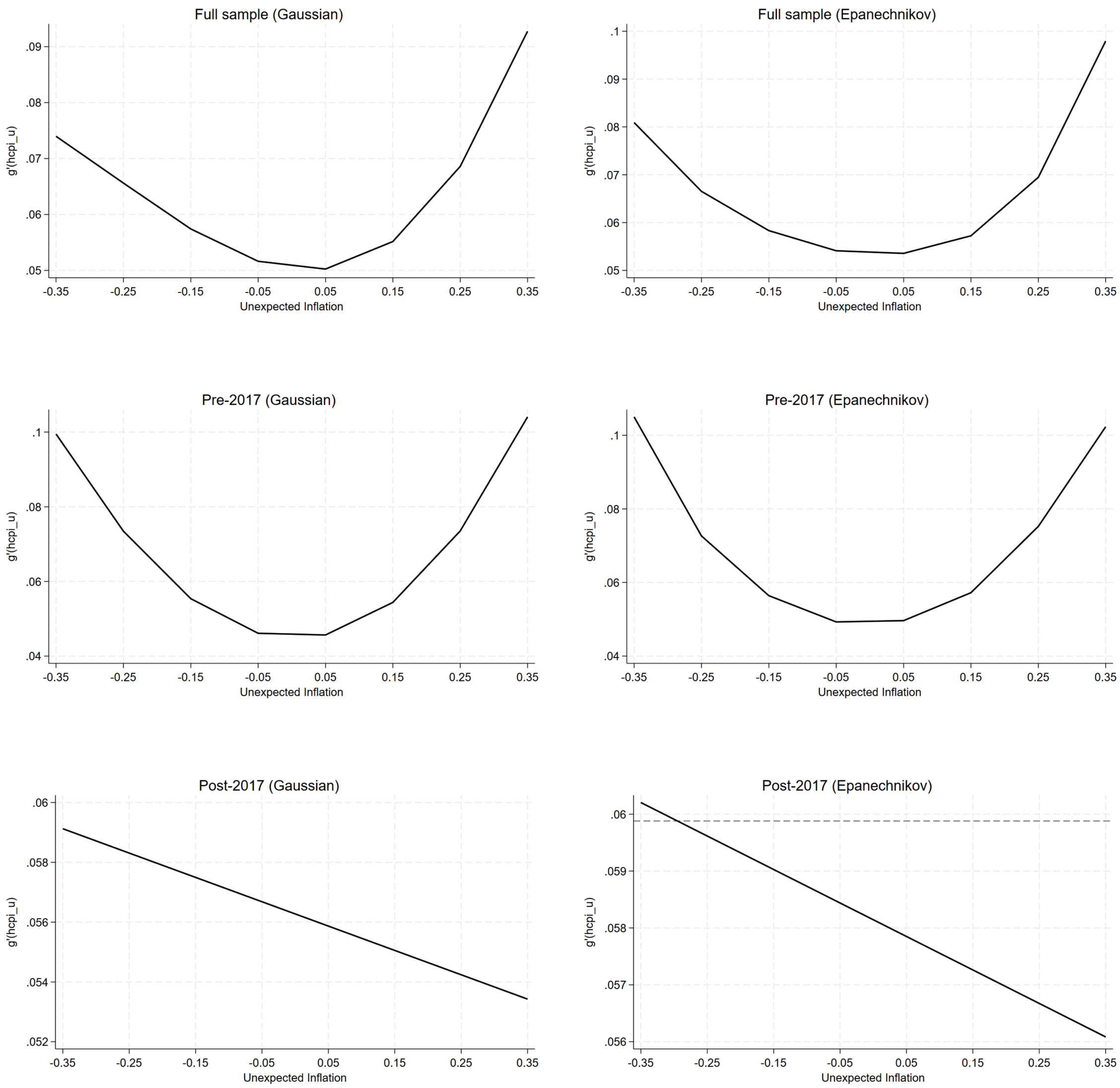

Figure 7: Semi-parametric regressions for unexpected inflation. Note: hcpi_u is the unexpected component of headline inflation.

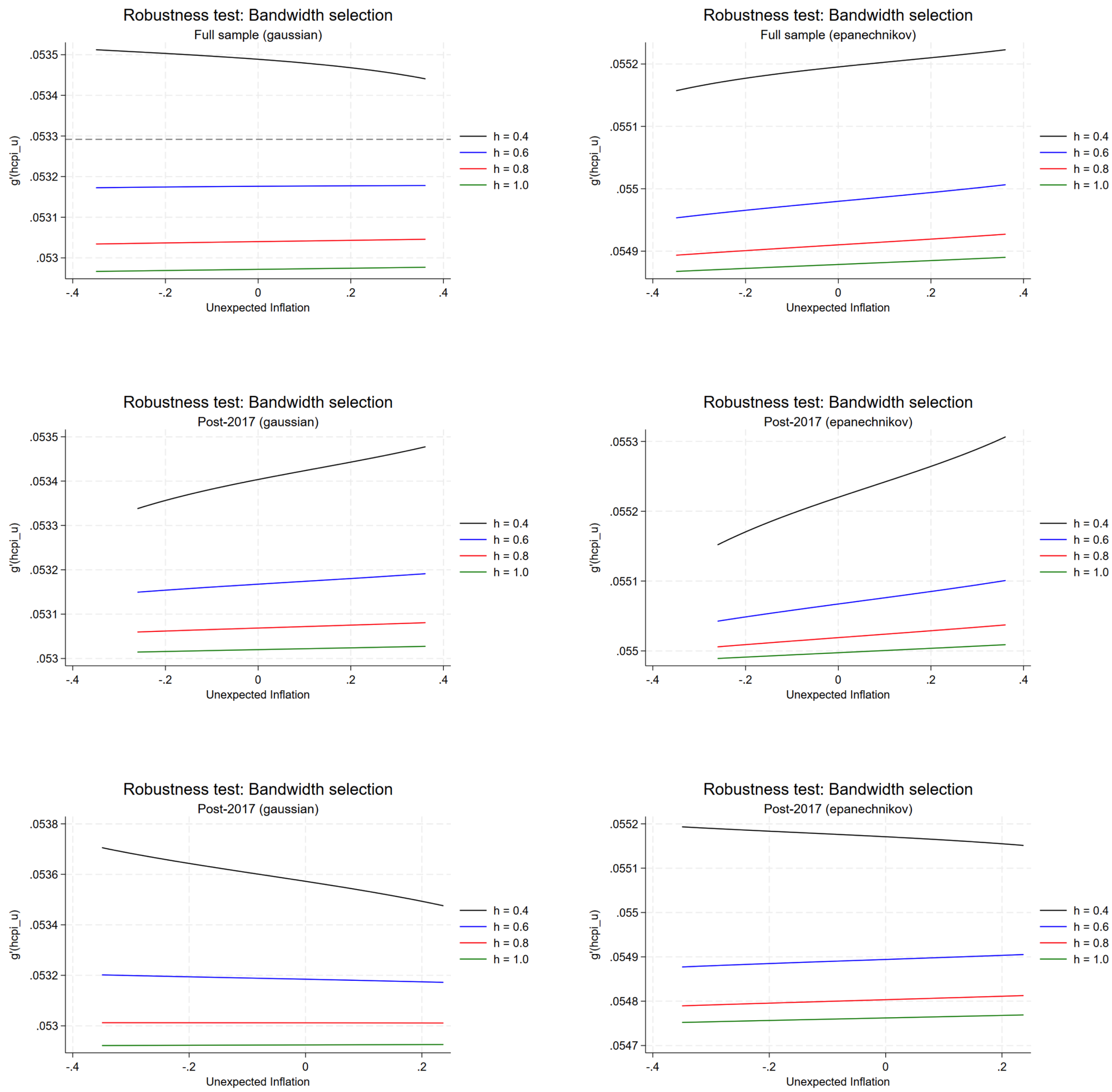

Figure 8: Semi-parametric regressions for unexpected inflation with alternative bandwidth selection. Note: hcpi_u is the unexpected component of headline inflation.

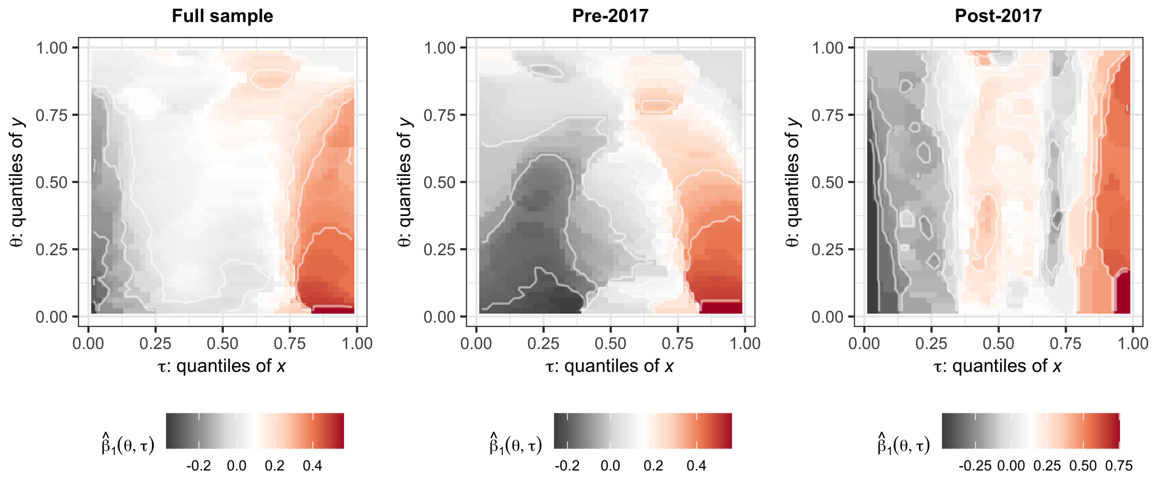

Figure 9: Quantile on quantile regressions for headline inflation. Note: In the quantile on quatile regression plots, the x-axis represents the quantiles of headline inflation, while the y-axis represents the quantiles of RPD. The color gradient indicates the strength and direction of the relationship between headline inflation and RPD across different quantiles. Warmer colors (e.g., red) indicate a stronger positive relationship, while cooler colors (e.g., blue) indicate a stronger negative relationship. The white areas represent quantiles where the relationship is not statistically significant.

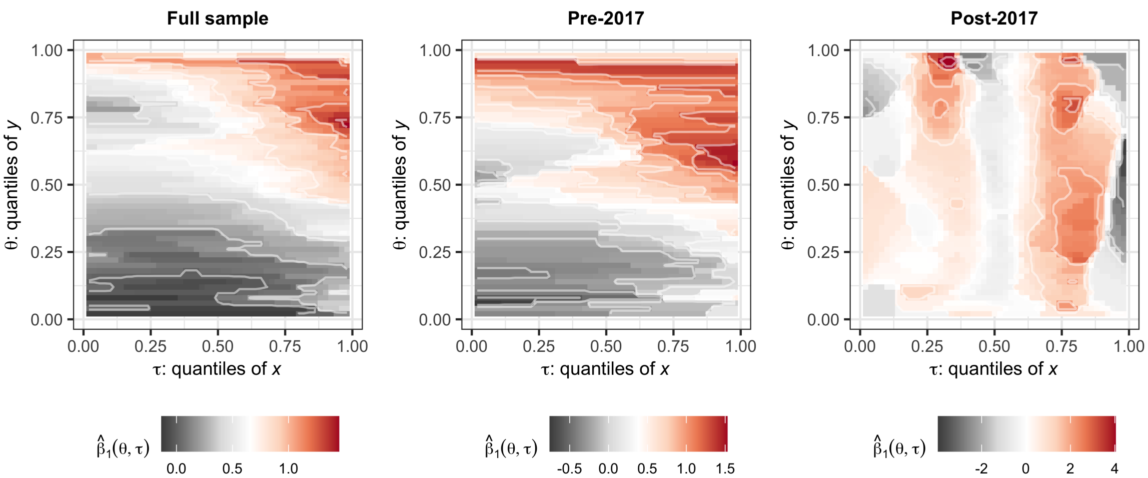

Figure 10: Quantile on quantile regressions for the unexpected componenet of inflation. Note: The x-axis represents the quantiles of the unexpected componenet of headline inflation, while the y-axis represents the quantiles of RPD.

7 Conclusion

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A Appendix: Data and descriptive statistics

A.1 Data

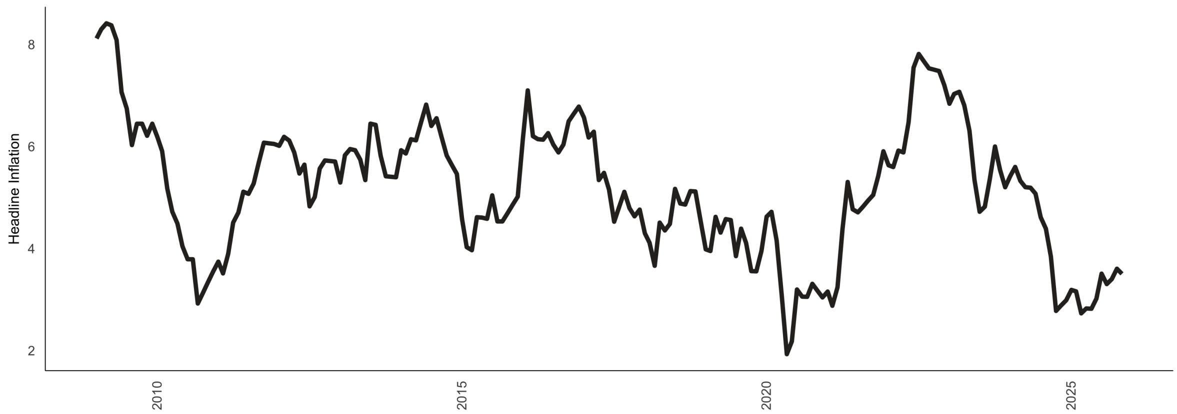

Figure A1: Headline inflation (%).

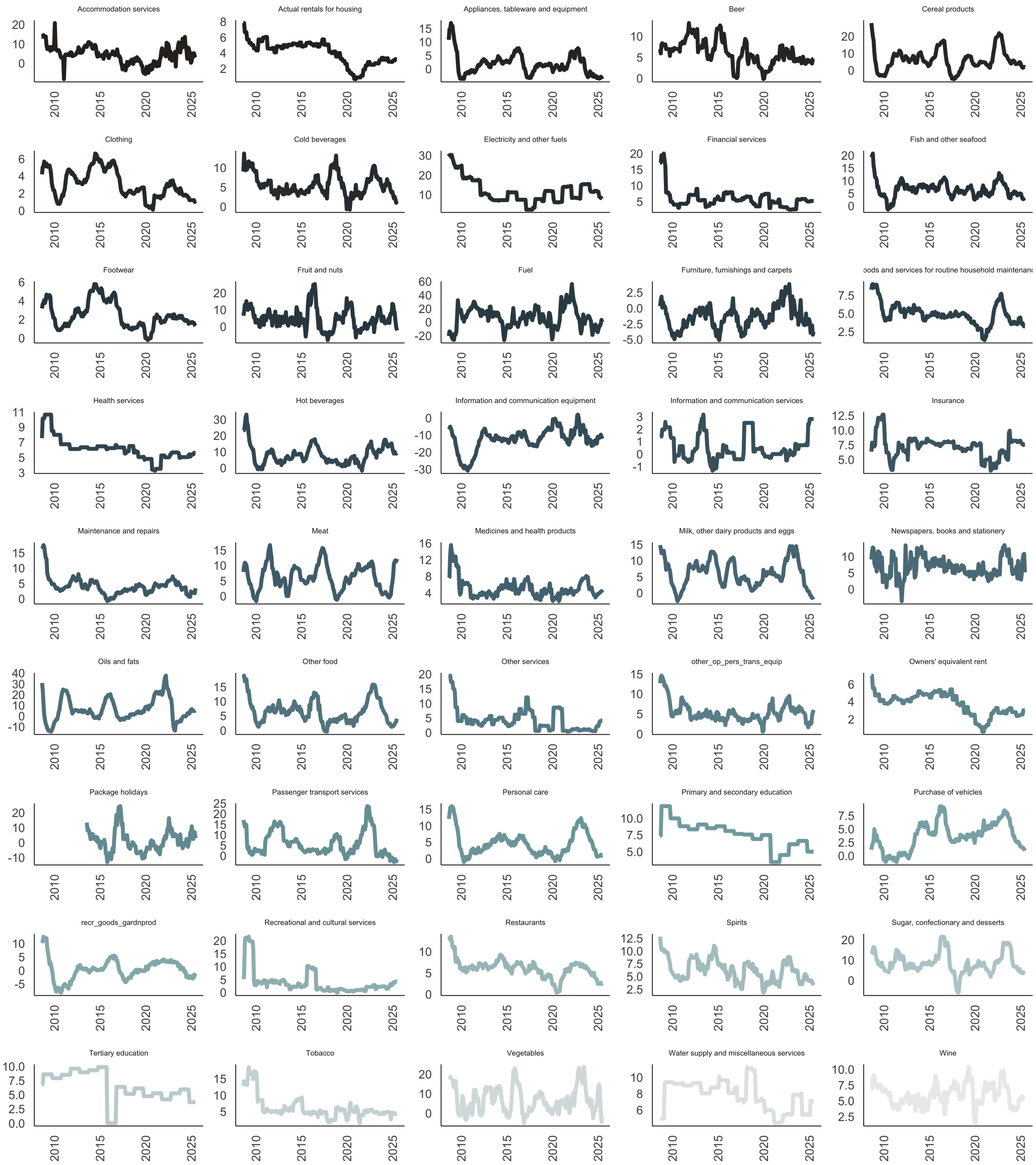

Figure A2: Inflation categories (%)

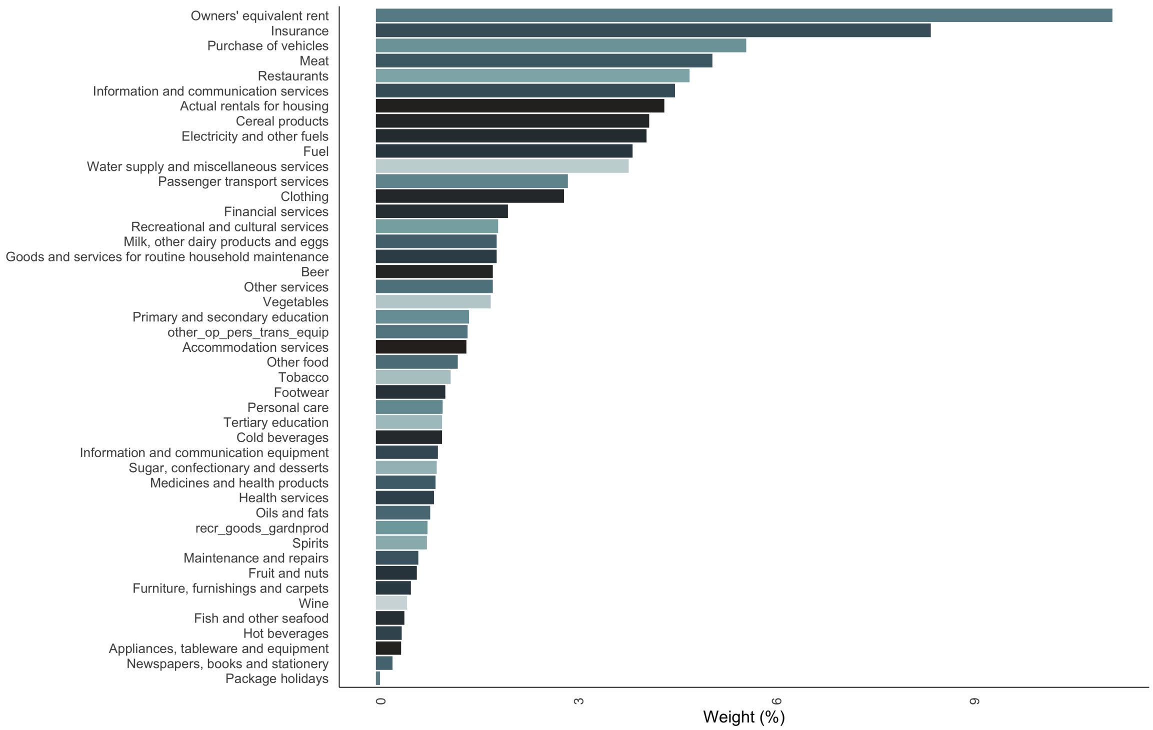

Figure A3: CPI Weights. Note: Due to a lack of data, some categories (no spitirits; transport services of goods; household textiles; glassware,tableware, and house utensils; and Tools and equipment for house and garden) have been excluded.

A.2 Descriptive statistics

Table A1: Descriptive statistics

Goods category	Mean	Median	SD	Min	Max	Observations
Accommodation services	4.47	4.55	4.43	-8.16	20.79	203
Actual rentals for housing	4.03	4.56	1.48	0.67	7.80	203
Appliances, tableware and equipment	1.81	1.50	3.93	-3.75	16.97	203
Beer	6.07	5.72	2.85	-0.25	13.19	203
Cereal products	6.19	5.17	6.17	-5.18	27.78	203
Clothing	3.06	2.68	1.63	0.11	6.60	203
Cold beverages	5.51	5.02	2.69	-0.76	13.59	203
Electricity and other fuels	11.78	10.66	6.19	2.28	30.99	203
Financial services	5.82	5.43	2.96	2.51	20.00	203
Fish and other seafood	6.54	6.43	3.22	-1.47	20.74	203
Footwear	2.45	2.16	1.40	-0.22	5.77	203
Fruit and nuts	4.75	4.25	6.16	-8.18	25.36	203
Fuel	6.51	7.33	15.01	-26.78	56.28	203
Furniture, furnishings and carpets	-1.42	-1.43	1.85	-5.11	3.83	203
Goods and services for routine household maintenance	4.88	4.76	1.44	1.32	9.03	203
Health services	6.12	6.11	1.48	3.26	10.65	203
Hot beverages	8.16	6.90	5.98	-1.70	33.08	203
Information and communication equipment	-12.55	-11.75	6.61	-30.92	1.93	203
Information and communication services	0.56	0.20	1.05	-1.32	3.15	203
Insurance	7.19	7.51	1.81	3.14	12.68	203
Maintenance and repairs	4.09	3.85	2.84	-0.80	17.53	203
Meat	6.20	6.03	4.16	-1.67	16.67	203
Medicines and health products	5.05	4.46	2.31	1.97	15.53	203
Milk, other dairy products and eggs	6.07	6.12	4.16	-2.60	14.83	203
Newspapers, books and stationery	6.81	6.56	3.03	-3.72	13.50	203
Oils and fats	6.17	4.21	10.55	-14.55	37.67	203
Other food	6.56	6.11	4.00	-0.45	18.83	203
Other services	4.28	3.40	3.83	0.31	19.47	203
Owners' equivalent rent	3.69	3.92	1.25	0.56	6.98	203
Package holidays	1.14	0.36	7.23	-13.17	24.37	203
Passenger transport services	5.67	3.15	5.75	-3.09	23.67	203
Personal care	4.51	3.86	3.68	-1.07	15.93	203
Primary and secondary education	7.60	7.64	1.98	3.42	11.80	203
Purchase of vehicles	3.66	3.80	2.60	-1.23	9.28	203
Recreational and cultural services	3.77	2.24	4.64	-0.11	21.58	203
Restaurants	6.03	6.18	2.16	0.38	13.50	203
Spirits	6.15	6.04	2.14	1.81	12.79	203
Sugar, confectionary and desserts	8.51	7.80	5.06	-6.03	21.61	203
Tertiary education	6.48	6.11	2.52	0.00	9.85	203
Tobacco	6.04	4.93	3.44	1.40	18.58	203
Vegetables	6.89	5.72	6.65	-5.14	23.59	203
Water supply and miscellaneous services	7.77	8.02	1.79	4.38	11.23	203
Wine	5.97	5.93	1.68	1.47	10.35	203
coreinfl	4.57	4.54	1.13	2.50	8.46	203
hinfl	5.13	5.17	1.32	1.93	8.41	203
other_op_pers_trans_equip	5.37	4.94	2.44	0.52	14.60	203
recr_goods_gardnprod	0.42	0.39	3.83	-8.23	12.51	203

A.3 Stationarity tests

Table A2: Stationarity tests results

Measure	ADF test (p-value)	PP test (p-value)	KPSS test (p-value)	Stationarity
Headline Inflation	0.14	0.20	0.04	Stationary
RPD	0.08	0.31	0.01	Stationary

A.4 Lag length selection

Table A3: RPD lag length selection

Tests	Optimal Lags
AIC(n)	3
HQ(n)	3
SC(n)	1
FPE(n)	3

B Appendix: Alternative Results (OLS Robustness)

The coefficient of variation (CV) version of RPD (or RPD-CV) normalizes the RPD by the aggregate price level to account for changes in the overall price level. This measure is particularly useful when comparing RPD across different time periods or economies with varying inflation rates. The RPD-CV is calculated as follows:

$$RPD - CV_t = \sqrt{\frac{\sum_{i=1}^{n} \omega_i (\pi_{i,t} - \pi_t)^2}{|1 + \pi_t|}}, \tag{11}$$

where π_t is the aggregate inflation rate at time t, and ω_i are the inflation basket weights.

B.1 RPD-CV over-time

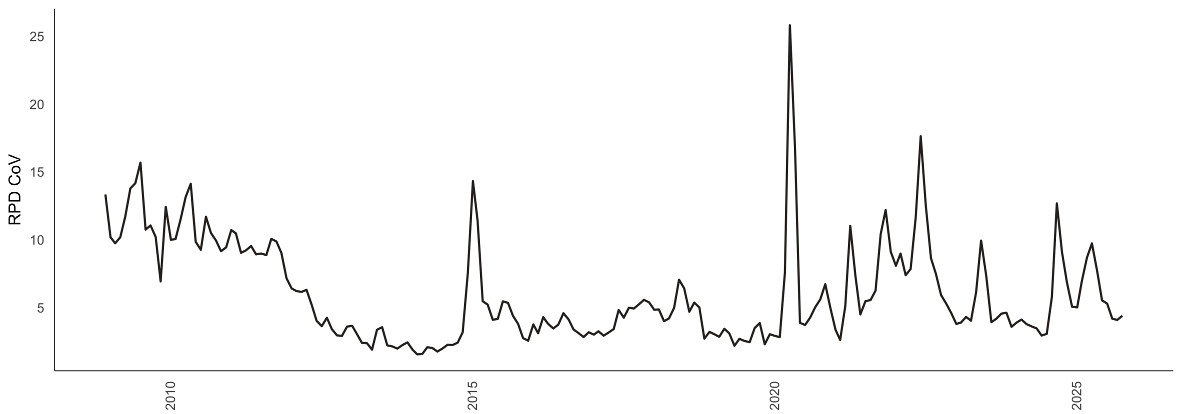

Figure B1: RPD-CV overtime

B.2 Stationarity tests for RPD-CV

Table B1: Stationarity tests results for RPD-CV

Measure	ADF test (p-value)	PP test (p-value)	KPSS test (p-value)	Stationarity
Headline Inflation	0.14	0.20	0.04	Stationary
RPD-CV	0.15	0.01	0.02	Stationary

B.3 Lag length selection for RPD-CV

Table B2: RPD-CV lag length selection

Tests	Optimal Lags
AIC(n)	3
HQ(n)	3
SC(n)	3
FPE(n)	3

B.4 RPD-CV baseline regression results

Table B3: RPD-CV baseline regressions

	Full Sample	Pre 2017	Post 2017	Full Sample: Squared Inflation	Pre 2017: Squared Inflation	Post 2017: Squared Inflation
π_t	-0.014	-0.006	0.001	-0.242**	-0.342*	-0.383**
π_t^2				0.023**	0.031*	0.04***
RPD_{t-3}	0.348***	0.154	0.361***	0.329***	0.126	0.286***
RPD_{t-2}	-0.648***	-0.298*	-0.792***	-0.618***	-0.259*	-0.725***
RPD_{t-1}	1.18***	1.077***	1.171***	1.145***	1.036***	1.065***
D_{2022m8}	-0.144*		-0.074	-0.235		-0.212
D_{2020m2}	-0.371***		-0.498*	-0.338		-0.45
D_{2017m7}	0.295***			0.314		
D_{2014m7}	-0.125**	-0.139		-0.159	-0.17	
$D_{2011m12}$	0.056	-0.009		0.083	0.028	
$Constant$	0.261**	0.132	0.43***	0.844***	1.066**	1.475***

Note: *** = p value < 0.01, ** = 0.01 < p value < 0.05, * = 0.05 < p value < 0.1.

C Appendix: Explanation of the quantile on quantile regression

To further capture the dynamic dependencies in the data, we perform a quantile on quantile regressions. Quantile regressions address the limitations of OLS which only estimates the mean dependency between on or more independent variables and the dependant variable. Distribution moments, such as the mean, can be strongly affected by heavy tails (or tail behaviour), non-linearity and extreme values in general (see Koenker, 2017). In response to these limitations, the quantile regression was first introduced by Koenker and Bassett Jr (1978). Koenker (2017) notes that quantile regressions are inherently local and are immune to small deviations in distributions.

However, Gupta et al. (2018) note that standard quantile regressions are limited in their ability to capture dependency in its entirety. That is, although quantile regressions capture the relationship between two variable (for example) at various points of the conditional distribution, it restricts the possibility that the nature of the independent variable can also influence how the independent variable is calculated. The quantile-on-quantile regression (QQR), therefore, offers a more complete picture of the complex dependency structure, and was utilised numerous times in the literature (see Mishra et al., 2019; Chang et al., 2020; Sim, 2016).

Given this brief explanation on QQR, we estimate the relationship between the θ-quantile of RPD (a_{2t}) and the τ-quantile of headline inflation (Γ_t) as follows:

$$a_{2t} = \beta^{\theta}\Gamma_t + \epsilon_t^{\theta}, \tag{12}$$

Taking the linear version of $\beta^{\theta}(.)$ with a first order Taylor expansion of $\beta^{\theta}(.)$ around Γ^{τ} gives:

$$\beta^{\theta}(\Gamma_t) \approx \beta^{\theta}(\Gamma^{\tau}) + \beta^{\theta'}(\Gamma^{\tau})(\Gamma_t - \Gamma^{\tau}) \tag{13}$$

In this instance both $\beta^{\theta}(\Gamma^{\tau})$ and $\beta^{\theta'}(\Gamma^{\tau})$ are indexed to θ and τ. This means that Equation 13 can be summarised as follows:

$$\beta^{\theta}(\Gamma_t) \approx \beta_0(\theta, \tau) + \beta_1(\theta, \tau)(\Gamma_t - \Gamma), \tag{14}$$

and collecting terms gives and substituting into Equation 12 results in :

$$a_{2t} = \beta_0(\theta,\tau) + \beta_1(\theta,\tau)(\Gamma^\tau)(\Gamma_t - \Gamma) + \epsilon_t^\theta \tag{15}$$

where $\beta^\theta(\Gamma^\tau)$ and $\beta^{\theta'}(\Gamma^\tau)$ are substituted with $\beta_0(\theta,\tau)$ and $\beta_1(\theta,\tau)(\Gamma^\tau)$ respectively.

Therefore, Equation 15 captures the relationship between the θ-quantile of the RPD and the τ-quantile of inflation, that is, the overall dependence structure of the respective distributions. To this end, to get the estimates of $\hat{\beta}_0(\theta,\tau)$ and $\hat{\beta}_1(\theta,\tau)$ we solve for:

$$\underset{b_0,b_1}{min} \sum_{i=1}^{n} \rho_\theta \left[a_{2t} - b_0 - b_1(\widehat{\Gamma_t} - \hat{\Gamma})\right] K\left(\frac{F_n(\widehat{\Gamma_t}) - \tau}{h}\right) \tag{16}$$

where ρ_θ is a tilted absolute value function which produces θ-quantile values of a_{2t}. A Gaussian kernel $K(.)$ is used to weight observations in the neighbourhood of $\widehat{\Gamma^\tau}$ using h as a bandwidth. These weights are inversely related to $\widehat{\Gamma_t} - \hat{\Gamma}$. Lastly, the choice of h remains uncertain in kernel regression where if a small h is chosen the bias of the estimates is smaller but the variance of these estimates increases, and conversely. To overcome this uncertainty we use the cross validation method to calculate the appropripate level of h.